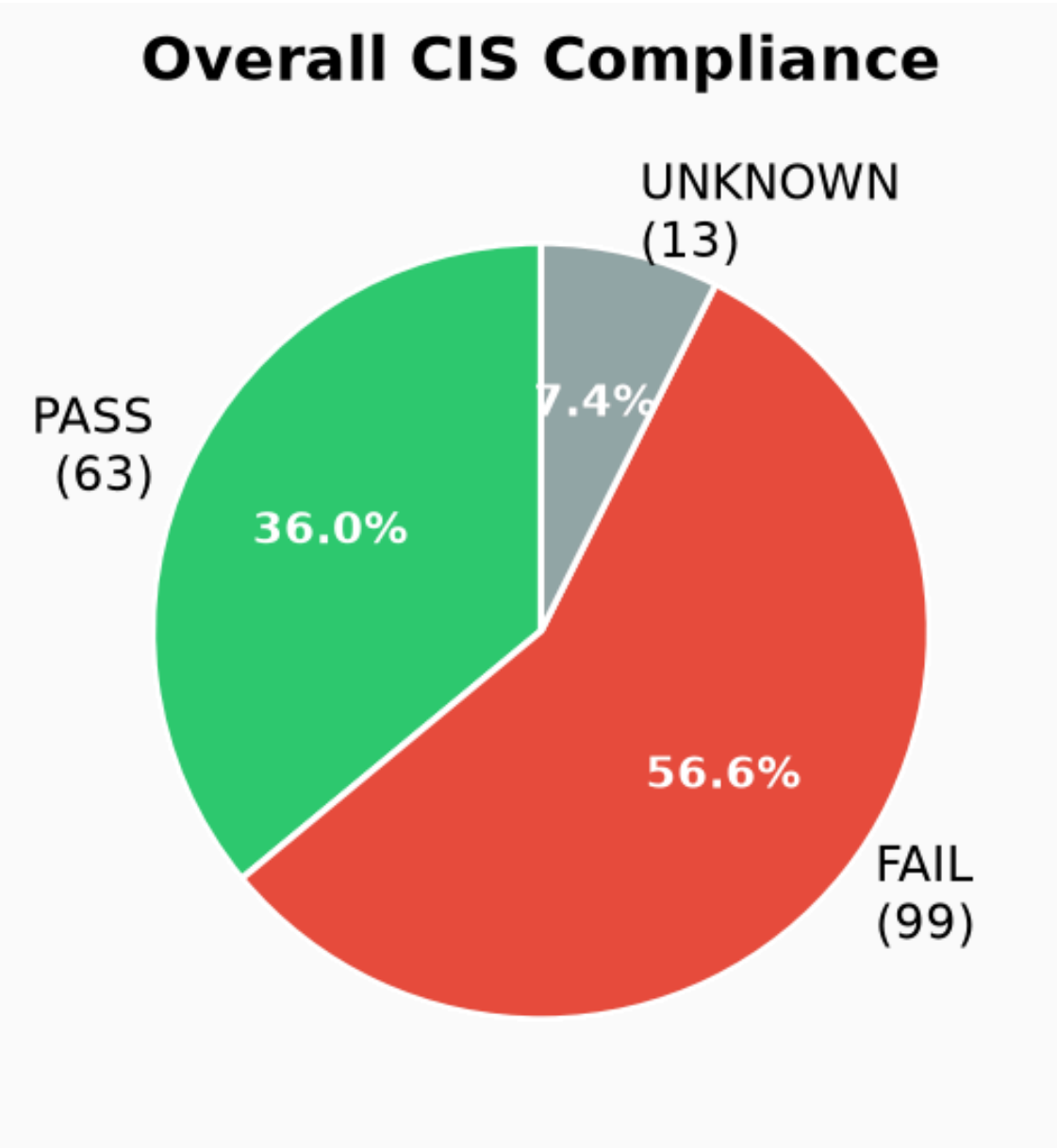


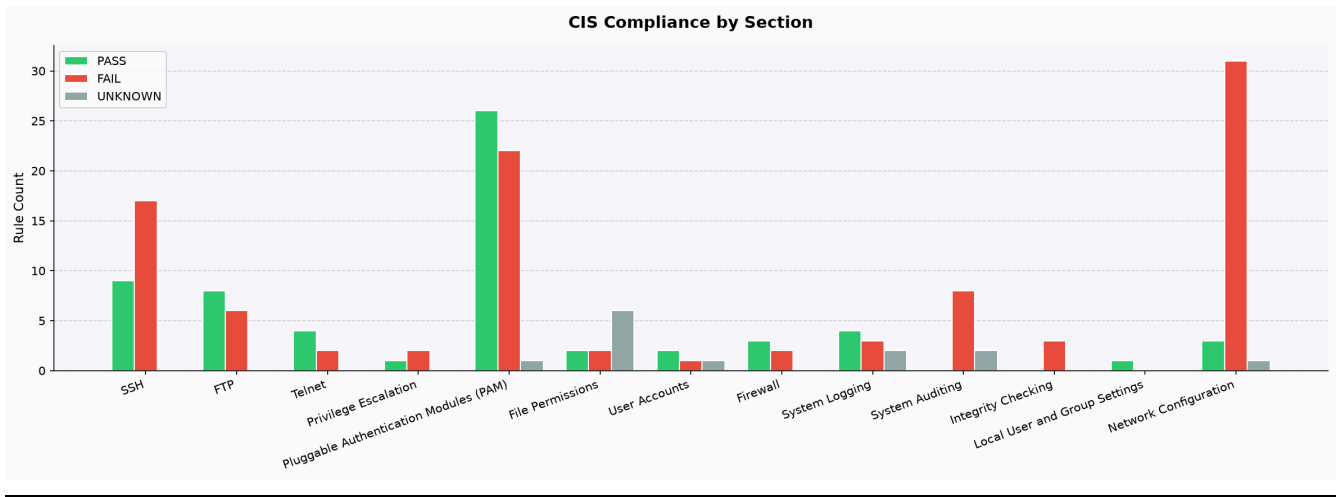
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	175
✓ Passed	63
✗ Failed	99
? Unknown	13
Overall Pass Rate	36.0%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
SSH	9	17	0	35%
FTP	8	6	0	57%
Telnet	4	2	0	67%
Privilege Escalation	1	2	0	33%
Pluggable Authentication Modules (PAM)	26	22	1	53%
File Permissions	2	2	6	20%
User Accounts	2	1	1	50%
Firewall	3	2	0	60%
System Logging	4	3	2	44%
System Auditing	0	8	2	0%
Integrity Checking	0	3	0	0%
Local User and Group Settings	1	0	0	100%
Network Configuration	3	31	1	9%

CIS Benchmark Compliance Report

Here is the security compliance report for the remote machine based on the provided CIS Ubuntu 24.04 LTS Benchmark rules and security profile.

SSH Rules

Rule 1.6.5: Ensure sshd warning Banner is configured

- **Status:** FAIL
- **Evidence:** `ssh.features.banner` is "none".
- **Recommendation:** Configure a warning banner in `/etc/ssh/sshd_config` by setting `Banner /etc/issue.net` or a similar file containing a warning message.

Rule 1.6.10: Ensure access to sshd warning banner is configured

- **Status:** FAIL
- **Evidence:** `ssh.features.banner` is "none".
- **Recommendation:** Configure a warning banner in `/etc/ssh/sshd_config` by setting `Banner /etc/issue.net` or a similar file containing a warning message. Ensure the banner file has appropriate permissions (e.g., 644 root:root).

Rule 5.1.1: Ensure access to `/etc/ssh/sshd_config` is configured

- **Status:** FAIL
- **Evidence:** `/etc/ssh/sshd_config` permissions are 644 root root.
- **Recommendation:** Set permissions on `/etc/ssh/sshd_config` to 0600 or 0640 to restrict access. For example: `chmod 600 /etc/ssh/sshd_config && chown root:root /etc/ssh/sshd_config`.

Rule 5.1.2: Ensure access to SSH private host key files is configured

- **Status:** PASS
- **Evidence:** SSH private host key files (e.g., `/etc/ssh/ssh_host_ecdsa_key`, `/etc/ssh/ssh_host_ed25519_key`, `/etc/ssh/ssh_host_rsa_key`) have permissions 600 root root.
- **Recommendation:** None

Rule 5.1.3: Ensure access to SSH public host key files is configured

- **Status:** PASS
- **Evidence:** SSH public host key files (e.g., `/etc/ssh/ssh_host_ecdsa_key.pub`, `/etc/ssh/ssh_host_ed25519_key.pub`, `/etc/ssh/ssh_host_rsa_key.pub`) have permissions 644 root root.
- **Recommendation:** None

Rule 5.1.4: Ensure sshd access is configured

- **Status:** FAIL
- **Evidence:** `ssh.authentication.permit_root_login` is set to "without-password".
- **Recommendation:** Set `PermitRootLogin no` in `/etc/ssh/sshd_config` to prevent direct root login.

Rule 5.1.5: Ensure sshd Banner is configured

- **Status:** FAIL
- **Evidence:** `ssh.features.banner` is "none".
- **Recommendation:** Configure a banner in `/etc/ssh/sshd_config` by setting `Banner /etc/issue.net` or a similar file containing a warning message.

Rule 5.1.6: Ensure sshd Ciphers are configured

- **Status:** PASS
- **Evidence:** Configured ciphers include `chacha20-poly1305@openssh.com`, `aes128-ctr`, `aes192-ctr`, `aes256-ctr`, `aes128-gcm@openssh.com`, `aes256-`

gcm@openssh.com. These are considered strong.

- **Recommendation:** None

Rule 5.1.7: Ensure sshd ClientAliveInterval and ClientAliveCountMax are configured

- **Status:** FAIL
- **Evidence:** `ssh.session.client_alive_interval` is 0 (disabled).
`ssh.session.client_alive_count_max` is 3.
- **Recommendation:** Set `ClientAliveInterval` to 300 (5 minutes) and `ClientAliveCountMax` to 0 or 3 in `/etc/ssh/sshd_config` to terminate idle sessions.

Rule 5.1.8: Ensure sshd DisableForwarding is enabled

- **Status:** FAIL
- **Evidence:** `ssh.forwarding.allow_tcp_forwarding` is true,
`ssh.forwarding.allow_agent_forwarding` is true,
`ssh.forwarding.allow_streamlocal_forwarding` is true.
- **Recommendation:** Set `AllowTcpForwarding` no, `AllowAgentForwarding` no, and `AllowStreamLocalForwarding` no in `/etc/ssh/sshd_config` to disable forwarding.

Rule 5.1.9: Ensure sshd GSSAPIAuthentication is disabled

- **Status:** PASS
- **Evidence:** `ssh.authentication.gssapiauthentication` is false.
- **Recommendation:** None

Rule 5.1.10: Ensure sshd HostbasedAuthentication is disabled

- **Status:** PASS
- **Evidence:** `ssh.authentication.hostbasedauthentication` is false.
- **Recommendation:** None

Rule 5.1.11: Ensure sshd IgnoreRhosts is enabled

- **Status:** PASS
- **Evidence:** `ssh.authentication.ignorerhosts` is true.
- **Recommendation:** None

Rule 5.1.12: Ensure sshd KexAlgorithms is configured

- **Status:** FAIL
- **Evidence:** `crypto.weak_kex` lists several weak algorithms, including `ecdh-sha2-nistp256`, `ecdh-sha2-nistp384`, `ecdh-sha2-nistp521`, and `diffie-hellman-group14-sha256`.
- **Recommendation:** Remove weak KexAlgorithms from the KexAlgorithms directive in `/etc/ssh/sshd_config`. Refer to the CIS Benchmark for recommended algorithms.

Rule 5.1.13: Ensure sshd LoginGraceTime is configured

- **Status:** FAIL
- **Evidence:** `ssh.session.login_grace_time` is 120 seconds.
- **Recommendation:** Set `LoginGraceTime` to 60 seconds or less in `/etc/ssh/sshd_config` to reduce the window for brute-force attacks.

Rule 5.1.14: Ensure sshd LogLevel is configured

- **Status:** FAIL
- **Evidence:** `ssh.features.log_level` is "INFO".
- **Recommendation:** Set `LogLevel VERBOSE` in `/etc/ssh/sshd_config` to ensure sufficient logging for auditing.

Rule 5.1.15: Ensure sshd MACs are configured

- **Status:** FAIL
- **Evidence:** `crypto.weak_macs` lists several weak algorithms, including `umac-64-etm@openssh.com`, `hmac-sha1-etm@openssh.com`, `umac-64@openssh.com`, `umac-128@openssh.com`, `hmac-sha2-256`, `hmac-sha2-512`, and `hmac-sha1`.
- **Recommendation:** Remove weak MAC algorithms from the `MACs` directive in `/etc/ssh/sshd_config`. Refer to the CIS Benchmark for recommended algorithms.

Rule 5.1.16: Ensure sshd MaxAuthTries is configured

- **Status:** FAIL
- **Evidence:** `ssh.authentication.max_auth_tries` is 6.
- **Recommendation:** Set `MaxAuthTries` to 4 or less in `/etc/ssh/sshd_config` to limit the number of authentication attempts.

Rule 5.1.17: Ensure sshd MaxStartups is configured

- **Status:** FAIL
- **Evidence:** `ssh.session.maxstartups` is "10:30:100".
- **Recommendation:** Set `MaxStartups` to "10:30:60" or less in `/etc/ssh/sshd_config` to limit concurrent unauthenticated connections.

Rule 5.1.18: Ensure sshd MaxSessions is configured

- **Status:** FAIL
- **Evidence:** `ssh.session.max_sessions` is 10.
- **Recommendation:** Set `MaxSessions` to 4 or less in `/etc/ssh/sshd_config` to limit the number of open sessions per network connection.

Rule 5.1.19: Ensure sshd PermitEmptyPasswords is disabled

- **Status:** PASS
- **Evidence:** `ssh.authentication.permit_empty_passwords` is false.
- **Recommendation:** None

Rule 5.1.20: Ensure sshd PermitRootLogin is disabled

- **Status:** FAIL
- **Evidence:** `ssh.authentication.permit_root_login` is set to "without-

password".

- **Recommendation:** Set `PermitRootLogin no` in `/etc/ssh/sshd_config` to prevent direct root login.

Rule 5.1.21: Ensure sshd PermitUserEnvironment is disabled

- **Status:** PASS
- **Evidence:** `ssh.features.permit_user_environment` is false.
- **Recommendation:** None

Rule 5.1.22: Ensure sshd UsePAM is enabled

- **Status:** PASS
- **Evidence:** `ssh.authentication.usepam` is true.
- **Recommendation:** None

Rule 5.1.23: Ensure sshd post-quantum cryptography key exchange algorithms are configured

- **Status:** FAIL
- **Evidence:** `crypto.weak_kex` lists several weak algorithms, including `ecdh-sha2-nistp256`, `ecdh-sha2-nistp384`, `ecdh-sha2-nistp521`, and `diffie-hellman-group14-sha256`. These are not considered post-quantum safe or are otherwise weak.
- **Recommendation:** Configure `KexAlgorithms` in `/etc/ssh/sshd_config` to use only strong, post-quantum safe algorithms as recommended by the CIS Benchmark.

Rule 5.1.24: Ensure sshd ListenAddress is configured

- **Status:** FAIL
- **Evidence:** `ssh.network.listen_address` is `"0.0.0.0:22"`, which listens on all interfaces.
- **Recommendation:** Set `ListenAddress` to a specific IP address (e.g., `ListenAddress 192.168.1.100:22`) in `/etc/ssh/sshd_config` to restrict SSH access to a designated network interface.

FTP Rules

Rule 2.1.8: Ensure ftp server services are not in use

- **Status:** FAIL
- **Evidence:** `ftp.service.installed` is true, `ftp.service.enabled` is true, and `ftp.service.running` is true for `vsftpd`.
- **Recommendation:** Uninstall the `vsftpd` package if FTP services are not required: `sudo apt purge vsftpd`. If FTP is required, ensure it is securely configured (e.g., disable anonymous access, enforce TLS, chroot users).

Rule 2.1.20: Ensure tftp server services are not in use

- **Status:** PASS
- **Evidence:** `ftp.tftp_server_packages` is "not installed" and `ftp.tftp_service_status` is "not running".

- **Recommendation:** None

Rule 2.2.6: Ensure ftp client is not installed

- **Status:** FAIL
- **Evidence:** The ftp client package is installed: `ii ftp 20230507-2build3 all dummy transitional package for tnftp.`
- **Recommendation:** Uninstall the ftp client package if not required: `sudo apt purge ftp.`

TELNET Rules

Rule 2.1.19: Ensure telnet server services are not in use

- **Status:** FAIL
- **Evidence:** `telnet.service.installed is true, telnet.network.port_open is true, and telnet.configuration.inetd_enabled is true.`
- **Recommendation:** Uninstall the telnetd package if Telnet services are not required: `sudo apt purge telnetd.` If inetd or xinetd is managing it, disable the telnet service in their configuration.

Rule 2.2.4: Ensure telnet client is not installed

- **Status:** FAIL
- **Evidence:** The telnet client package is installed: `ii telnet 0.17+2.5-3ubuntu4.2 all transitional dummy package for inetutils-telnet default switch.`
- **Recommendation:** Uninstall the telnet client package if not required: `sudo apt purge telnet.`

ESCALATION Rules

Rule 5.2.1: Ensure sudo is installed

- **Status:** PASS
- **Evidence:** `privilege_escalation.sudo_installed.installed is true with version 1.9.15p5.`
- **Recommendation:** None

Rule 5.2.2: Ensure sudo commands use pty

- **Status:** PASS
- **Evidence:** `privilege_escalation.use_pty_entries contains Defaults use_pty in /etc/sudoers.`
- **Recommendation:** None

Rule 5.2.3: Ensure sudo log file exists

- **Status:** PASS
- **Evidence:** `privilege_escalation.sudo_syslog_logging.auth_log_exists is true and syslog_log_exists is true.`

- **Recommendation:** None

Rule 5.2.4: Ensure users must provide password for escalation

- **Status:** PASS
- **Evidence:** `privilege_escalation.nopasswd_entries` is empty, indicating no NOPASSWD entries are configured.
- **Recommendation:** None

Rule 5.2.5: Ensure re-authentication for privilege escalation is not disabled globally

- **Status:** PASS
- **Evidence:** `privilege_escalation.noauthenticate_entries` is empty, indicating no NOAUTHENTICATE entries are configured.
- **Recommendation:** None

Rule 5.2.6: Ensure sudo timestamp_timeout is configured

- **Status:** FAIL
- **Evidence:** `privilege_escalation.timestamp_timeout_entries` is empty, indicating `timestamp_timeout` is not explicitly configured.
- **Recommendation:** Add `Defaults timestamp_timeout=X` (where X is a reasonable timeout in minutes, e.g., 0 for always prompt, 15 for 15 minutes) to `/etc/sudoers` or a file in `/etc/sudoers.d/`.

Rule 5.2.7: Ensure access to the su command is restricted

- **Status:** FAIL
- **Evidence:** The `pam_wheel.so` line in `/etc/pam.d/su` is commented out: `# auth required pam_wheel.so`.
- **Recommendation:** Uncomment `auth required pam_wheel.so` in `/etc/pam.d/su` and ensure that only authorized users are members of the `wheel` group.

DIRECTORY ACCESS Rules

Rule 7.1.1: Ensure access to /etc/passwd is configured

- **Status:** PASS
- **Evidence:** `/etc/passwd` has permissions `0644 root root`.
- **Recommendation:** None

Rule 7.1.2: Ensure access to /etc/passwd- is configured

- **Status:** PASS
- **Evidence:** `/etc/passwd-` has permissions `0644 root root`.
- **Recommendation:** None

Rule 7.1.3: Ensure access to /etc/group is configured

- **Status:** PASS

- **Evidence:** `/etc/group` has permissions `0644 root root`.
- **Recommendation:** None

Rule 7.1.4: Ensure access to `/etc/group`- is configured

- **Status:** PASS
- **Evidence:** `/etc/group-` has permissions `0644 root root`.
- **Recommendation:** None

Rule 7.1.5: Ensure access to `/etc/shadow` is configured

- **Status:** PASS
- **Evidence:** `/etc/shadow` has permissions `0640 root shadow`.
- **Recommendation:** None

Rule 7.1.6: Ensure access to `/etc/shadow-` is configured

- **Status:** PASS
- **Evidence:** `/etc/shadow-` has permissions `0640 root shadow`.
- **Recommendation:** None

Rule 7.1.7: Ensure access to `/etc/gshadow` is configured

- **Status:** PASS
- **Evidence:** `/etc/gshadow` has permissions `0640 root shadow`.
- **Recommendation:** None

Rule 7.1.8: Ensure access to `/etc/gshadow-` is configured

- **Status:** PASS
- **Evidence:** `/etc/gshadow-` has permissions `0640 root shadow`.
- **Recommendation:** None

Rule 7.1.9: Ensure access to `/etc/shells` is configured

- **Status:** PASS
- **Evidence:** `/etc/shells` has permissions `0644 root root`.
- **Recommendation:** None

Rule 7.1.10: Ensure access to `/etc/security/opasswd` is configured

- **Status:** PASS
- **Evidence:** `/etc/security/opasswd` has permissions `0600 root root`.
- **Recommendation:** None

Rule 7.1.11: Ensure world writable files and directories are secured

- **Status:** PASS
- **Evidence:** `file_permissions.world_writable.world_writable_files` and `file_permissions.world_writable.world_writable_dirs_no_sticky` are both empty.
- **Recommendation:** None

Rule 7.1.12: Ensure no files or directories without an owner and a group exist

- **Status:** PASS
- **Evidence:** `file_permissions.unowned.unowned_or_ungrouped` is empty.
- **Recommendation:** None

Rule 7.1.13: Ensure SUID and SGID files are reviewed

- **Status:** INFORMATIONAL
- **Evidence:** The system contains SUID/SGID files such as `/usr/bin/umount`, `/usr/bin/sudo`, `/usr/bin/su`, `/usr/bin/passwd`, `/usr/bin/chage`, `/usr/bin/crontab`, `/usr/bin/mount`, `/usr/lib/dbus-1.0/dbus-daemon-launch-helper`, `/usr/lib/x86_64-linux-gnu/utempter/utempter`, `/usr/lib/openssh/ssh-keysign`, `/usr/lib/polkit-1/polkit-agent-helper-1`, `/usr/sbin/pam_extrausers_chkpwd`, and `/usr/sbin/unix_chkpwd`.
- **Recommendation:** Regularly review all SUID/SGID files on the system to ensure they are authorized and necessary. Remove or restrict any unnecessary SUID/SGID binaries.

USER ACCOUNTS Rules

Rule 5.4.1.1: Ensure password expiration is configured

- **Status:** FAIL
- **Evidence:**
`user_accounts.shadow_password_suite.login_defs.PASS_MAX_DAYS` is 99999.
- **Recommendation:** Set `PASS_MAX_DAYS` to 365 or less in `/etc/login.defs` to enforce regular password changes.

Rule 5.4.1.2: Ensure minimum password days is configured

- **Status:** INFORMATIONAL
- **Evidence:**
`user_accounts.shadow_password_suite.login_defs.PASS_MIN_DAYS` is 0.
- **Recommendation:** CIS recommends setting `PASS_MIN_DAYS` to 1 or more to prevent immediate password changes. This is a manual review item.

Rule 5.4.1.3: Ensure password expiration warning days is configured

- **Status:** PASS
- **Evidence:**
`user_accounts.shadow_password_suite.login_defs.PASS_WARN_AGE` is 7.
- **Recommendation:** None

Rule 5.4.1.4: Ensure strong password hashing algorithm is configured

- **Status:** PASS
- **Evidence:** `user_accounts.shadow_password_suite.pam_unix_hashing` shows `yescrypt` is used.
- **Recommendation:** None

Rule 5.4.1.5: Ensure inactive password lock is configured

- **Status:** FAIL
- **Evidence:** `user_accounts.shadow_password_suite.useradd_inactive` is -1 (disabled).
- **Recommendation:** Set INACTIVE to 30 (or another appropriate value) in `/etc/default/useradd` and ensure INACTIVE is set for all users using `chage -I 30 <user>`.

Rule 5.4.1.6: Ensure all users last password change date is in the past

- **Status:** FAIL
- **Evidence:** Several user accounts, including `root`, `daemon`, `bin`, `sys`, `sync`, `games`, `man`, `lp`, `mail`, `news`, `uucp`, `proxy`, `www-data`, `backup`, `list`, `irc`, `_apt`, `nobody`, `systemd-network`, `systemd-timesync`, `dhcpcd`, `messagebus`, `systemd-resolve`, `pollinate`, `polkitd`, `syslog`, `uidd`, `tcpdump`, `tss`, `landscape`, `fwupd-refresh`, `usbmux`, `wazuh-indexer`, `wazuh`, `wazuh-dashboard`, `sshd`, and `ftp` have a `last_password_change` date in the future (Feb 16, 2025 or Apr 19, 2026 or Jun 25, 2026).
- **Recommendation:** For all accounts with future password change dates, set the last password change date to a past date using `chage -d $(date +%Y-%m-%d) <username>`.

Rule 5.4.2.1: Ensure root is the only UID 0 account

- **Status:** PASS
- **Evidence:** `user_accounts.root_and_system_accounts.uid_0_accounts` lists only "root".
- **Recommendation:** None

Rule 5.4.2.2: Ensure root is the only GID 0 account

- **Status:** PASS
- **Evidence:** `user_accounts.root_and_system_accounts.primary_gid_0_accounts` lists only "root".
- **Recommendation:** None

Rule 5.4.2.3: Ensure group root is the only GID 0 group

- **Status:** PASS
- **Evidence:** `user_accounts.root_and_system_accounts.gid_0_groups` lists only "root".
- **Recommendation:** None

Rule 5.4.2.4: Ensure root account access is controlled

- **Status:** FAIL
- **Evidence:** `user_accounts.root_and_system_accounts.securetty_exists` is false.
- **Recommendation:** Create and configure `/etc/securetty` to list only secure TTYs from which root may log in directly.

Rule 5.4.2.5: Ensure root path integrity

- **Status:** FAIL
- **Evidence:** `user_accounts.root_and_system_accounts.root_path` includes `/snap/bin`.
- **Recommendation:** Ensure the root user's PATH environment variable (`/root/.bashrc`, `/root/.bash_profile`, `/etc/profile`, `/etc/bash.bashrc`) does not contain world-writable directories or directories that are not owned by root. Remove `/snap/bin` if it's not strictly necessary and properly secured.

Rule 5.4.2.6: Ensure root user umask is configured

- **Status:** PASS
- **Evidence:** `user_accounts.root_and_system_accounts.root_umask.login_defs` is `"022"`.
- **Recommendation:** None

Rule 5.4.2.7: Ensure system accounts do not have a valid login shell

- **Status:** FAIL
- **Evidence:** The `sync` account has `/bin/sync` as its shell, which is a valid login shell.
- **Recommendation:** Change the shell for the `sync` account to `/usr/sbin/nologin` or `/bin/false` using `usermod -s /usr/sbin/nologin sync`.

Rule 5.4.2.8: Ensure accounts without a valid login shell are locked

- **Status:** PASS
- **Evidence:** All system accounts with `nologin` or `false` shells have `passwd_status` `"L"`, indicating they are locked.
- **Recommendation:** None

Rule 5.4.3.1: Ensure nologin is not listed in /etc/shells

- **Status:** PASS
- **Evidence:** `user_accounts.default_user_environment.nologin_in_shells` is `false`.
- **Recommendation:** None

Rule 5.4.3.2: Ensure default user shell timeout is configured

- **Status:** FAIL
- **Evidence:** `user_accounts.default_user_environment.tmout_settings` is empty, indicating no TMOUT setting is configured.
- **Recommendation:** Add `TMOUT=600` (or another appropriate value in seconds) to `/etc/profile.d/` or `/etc/bash.bashrc` to automatically log out idle shell sessions.

Rule 5.4.3.3: Ensure default user umask is configured

- **Status:** PASS
- **Evidence:** `user_accounts.default_user_environment.default_user_umask.login_def`

is "022".

- **Recommendation:** None

FIREWALL Rules

Rule 4.1.1: Ensure ufw is installed

- **Status:** PASS
- **Evidence:** `ufw_firewall.ufw_installed` is `true` and `ufw_firewall.ufw_dpkg_status` is `"install ok installed"`.
- **Recommendation:** None

Rule 4.1.2: Ensure ufw service is configured

- **Status:** FAIL
- **Evidence:** `ufw_firewall.service_configured.is_enabled` is `"enabled"` and `is_active` is `"active"`, but `status_verbose` is `"inactive"`. This indicates a discrepancy where the service is enabled but not actively running or enforcing rules.
- **Recommendation:** Ensure UFW is active and enforcing rules. Run `sudo ufw enable` and verify its status with `sudo ufw status verbose`.

Rule 4.1.3: Ensure ufw incoming default is configured

- **Status:** PASS
- **Evidence:** `ufw_firewall.default_policies.etc_default_ufw.DEFAULT_INPUT_POLICY` is `"DROP"`.
- **Recommendation:** None

Rule 4.1.4: Ensure ufw outgoing default is configured

- **Status:** FAIL
- **Evidence:** `ufw_firewall.default_policies.etc_default_ufw.DEFAULT_OUTPUT_POLICY` is `"ACCEPT"`.
- **Recommendation:** Set `DEFAULT_OUTPUT_POLICY` to `"DROP"` in `/etc/default/ufw` and configure explicit allow rules for necessary outbound traffic.

Rule 4.1.5: Ensure ufw routed default is configured

- **Status:** PASS
- **Evidence:** `ufw_firewall.default_policies.etc_default_ufw.DEFAULT_FORWARD_POLICY` is `"DROP"`.
- **Recommendation:** None

SYSTEM LOGGING Rules

Rule 6.1.1.1.1: Ensure systemd-journald service is enabled and active

- **Status:** PASS

- **Evidence:** `system_logging.journald.systemd-journald_active` is "active" and `system_logging.journald.systemd-journald_enabled` is "static".
- **Recommendation:** None

Rule 6.1.1.1.2: Ensure systemd-journal-remote service is not in use

- **Status:** PASS
- **Evidence:** `system_logging.journald.systemd-journal-remote_installed` is false.
- **Recommendation:** None

Rule 6.1.1.1.3: Ensure journald is configured to send logs to rsyslog

- **Status:** FAIL
- **Evidence:** The `ForwardToSyslog` option in `/etc/systemd/journald.conf` is commented out, implying it's not explicitly configured to forward logs to rsyslog.
- **Recommendation:** Uncomment and set `ForwardToSyslog=yes` in `/etc/systemd/journald.conf` to ensure journald logs are forwarded to rsyslog.

Rule 6.1.1.1.4: Ensure journald log file access is configured

- **Status:** INFORMATIONAL
- **Evidence:** Journald log directories `/run/log/journal` and `/var/log/journal` have permissions `2755 root journal`. Individual journal files have `0640 root journal`.
- **Recommendation:** Manually verify that these permissions are appropriate for your environment and meet security requirements.

Rule 6.1.1.1.5: Ensure journald log file rotation is configured

- **Status:** INFORMATIONAL
- **Evidence:** The `MaxRetentionSec` and `MaxFileSec` options in `/etc/systemd/journald.conf` are commented out, indicating default rotation settings are in use.
- **Recommendation:** Manually verify that the default journald log rotation settings are sufficient for your retention policies. Consider explicitly setting `SystemMaxUse`, `SystemKeepFree`, `SystemMaxFileSize`, or `MaxRetentionSec` in `/etc/systemd/journald.conf`.

Rule 6.1.1.1.6: Ensure journald Storage is configured

- **Status:** FAIL
- **Evidence:** The `Storage` option in `/etc/systemd/journald.conf` is commented out, implying `Storage=auto`.
- **Recommendation:** Uncomment and set `Storage=persistent` in `/etc/systemd/journald.conf` to ensure journald logs are stored persistently across reboots.

Rule 6.1.1.1.7: Ensure journald Compress is configured

- **Status:** FAIL
- **Evidence:** The `Compress` option in `/etc/systemd/journald.conf` is commented out, implying default compression settings.
- **Recommendation:** Uncomment and set `Compress=yes` in `/etc/systemd/`

`journald.conf` to ensure `journald` logs are compressed.

Rule 6.1.2.1: Ensure rsyslog is installed

- **Status:** PASS
- **Evidence:** `system_logging.rsyslog.rsyslog_installed` is `true`.
- **Recommendation:** None

Rule 6.1.2.2: Ensure rsyslog service is enabled and active

- **Status:** PASS
- **Evidence:** `system_logging.rsyslog.rsyslog_active` is "active" and `system_logging.rsyslog.rsyslog_enabled` is "enabled".
- **Recommendation:** None

Rule 6.1.2.3: Ensure rsyslog default file permissions are configured

- **Status:** PASS
- **Evidence:** `/etc/rsyslog.conf` sets `$FileOwner syslog`, `$FileGroup adm`, and `$FileCreateMode 0640`.
- **Recommendation:** None

Rule 6.1.2.4: Ensure rsyslog logging is configured

- **Status:** INFORMATIONAL
- **Evidence:** `/etc/rsyslog.d/50-default.conf` contains standard logging rules for `auth`, `authpriv.*`, `kern.*`, `mail.*`, etc.
- **Recommendation:** Manually review the rsyslog configuration files (`/etc/rsyslog.conf` and files in `/etc/rsyslog.d/`) to ensure all critical system events are logged as required by your security policy.

Rule 6.1.2.5: Ensure rsyslog is configured to send logs to a remote log host

- **Status:** INFORMATIONAL
- **Evidence:** The rsyslog configuration does not explicitly show remote log forwarding rules.
- **Recommendation:** If remote logging is required, configure rsyslog to send logs to a central log host using a secure protocol (e.g., TLS). Add a line like `*.* @remote-log-host:514 (UDP)` or `*.* @@remote-log-host:514 (TCP)` or use `omfwd` module with TLS.

Rule 6.1.2.6: Ensure rsyslog is not configured to receive logs from a remote client

- **Status:** PASS
- **Evidence:** The `imudp` and `imtcp` modules are commented out in `/etc/rsyslog.conf`, indicating rsyslog is not configured to receive logs from remote clients.
- **Recommendation:** None

Rule 6.1.2.7: Ensure logrotate is configured

- **Status:** INFORMATIONAL
- **Evidence:** logrotate is installed, its timer is active and enabled, and various log rotation configurations exist in `/etc/logrotate.conf` and `/etc/logrotate.d/`.
- **Recommendation:** Manually review the logrotate configuration files to ensure all critical log files are rotated, compressed, and retained according to your organization's policy.

Rule 6.1.2.8: Ensure rsyslog-gnutls is installed

- **Status:** FAIL
- **Evidence:** `system_logging.rsyslog.rsyslog_gnutls_installed` is false.
- **Recommendation:** Install the rsyslog-gnutls package: `sudo apt install rsyslog-gnutls`.

Rule 6.1.2.9: Ensure rsyslog is configured to use gtls for forwarding

- **Status:** FAIL
- **Evidence:** rsyslog-gnutls is not installed, so gtlis cannot be configured for forwarding.
- **Recommendation:** Install rsyslog-gnutls and then configure rsyslog to use GTLS for secure remote log forwarding.

Rule 6.1.2.10: Ensure rsyslog CA certificates are configured

- **Status:** INFORMATIONAL
- **Evidence:** The profile does not contain information about rsyslog CA certificate configuration.
- **Recommendation:** If rsyslog is configured for secure remote logging with TLS, ensure that appropriate CA certificates are configured and managed for authentication.

Rule 6.1.3.1: Ensure access to all logfiles has been configured

- **Status:** INFORMATIONAL
- **Evidence:** Various log files in `/var/log` have permissions ranging from 0600 to 0664. For example, `/var/log/auth.log` is 0640 `syslog adm`, `/var/log/syslog` is 0640 `syslog adm`, `/var/log/wtmp` is 0664 `root utmp`.
- **Recommendation:** Manually review the permissions of all log files to ensure they are strictly controlled and prevent unauthorized access. Adjust permissions as necessary to 0640 or more restrictive.

SYSTEM AUDITING (auditd) Rules

Rule 6.2.1.1.1: Ensure auditd is installed

- **Status:** FAIL
- **Evidence:** `system_auditing.auditd_installed` is false.
- **Recommendation:** Install the auditd package: `sudo apt install auditd`.

Rule 6.2.1.1.2: Ensure auditd service is enabled and active

- **Status:** FAIL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install and enable the auditd service: `sudo apt install auditd && sudo systemctl enable auditd && sudo systemctl start auditd.`

Rule 6.2.1.1.3: Ensure auditing for processes that start prior to auditd is enabled

- **Status:** FAIL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and ensure the kernel parameter `audit=1` is set in GRUB to enable auditing from boot.

Rule 6.2.1.1.4: Ensure audit_backlog_limit is sufficient

- **Status:** FAIL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and configure `audit_backlog_limit` in `/etc/audit/auditd.conf` to a value of 8192 or greater.

Rule 6.2.1.2.1: Ensure audit log storage size is configured

- **Status:** FAIL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and configure `max_log_file` in `/etc/audit/auditd.conf` to an appropriate size (e.g., 50 MB).

Rule 6.2.1.2.2: Ensure audit logs are not automatically deleted

- **Status:** FAIL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and set `max_log_file_action = keep_logs` in `/etc/audit/auditd.conf`.

Rule 6.2.1.2.3: Ensure system is disabled when audit logs are full

- **Status:** FAIL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and set `space_left_action = SYSLOG`, `action_mail_acct = root`, `admin_space_left_action = HALT` in `/etc/audit/auditd.conf`.

Rule 6.2.1.2.4: Ensure system warns when audit logs are low on space

- **Status:** FAIL
- **Evidence:** auditd is not installed.
- **Recommendation:** Install auditd and set `space_left_action = SYSLOG` and `admin_space_left_action = SYSLOG (or SINGLE or HALT)` in `/etc/audit/auditd.conf`.

Rule 6.2.3.1-6.2.3.30: Ensure auditd rules cover sudoers, identity, logins,

session, time-change, MAC policy, access, mounts, deletion, loading modules, and configuration is immutable

- **Status:** UNKNOWN
- **Evidence:** The security profile does not contain the `audit_rules_raw` object, which is necessary to verify the presence and correctness of specific auditd rules.
- **Recommendation:** Install `auditd` and configure a comprehensive set of audit rules as specified in the CIS Benchmark for Ubuntu 24.04 LTS. Ensure the rules are loaded and immutable (`-e 2`).

Rule 6.2.4.1-6.2.4.10: Ensure audit log files and audit tools file access and permissions are strictly constrained

- **Status:** UNKNOWN
- **Evidence:** The security profile does not contain the `audit_rules_raw` object or specific checks for audit log/tool file permissions.
- **Recommendation:** Install `auditd` and ensure that audit log files (`/var/log/audit/`) and audit tools (`/sbin/auditctl`, `/sbin/auditd`, etc.) have appropriate restrictive permissions (e.g., `0640` for logs, `0755` for tools) and are owned by `root:root` or `root:adm`.

Integrity Checking

Rule 6.3.1: Ensure AIDE is installed

- **Status:** FAIL
- **Evidence:** `aide_integrity_checking.aide_installed` is false.
- **Recommendation:** Install the AIDE package: `sudo apt install aide aide-common`.

Rule 6.3.2: Ensure filesystem integrity is regularly checked

- **Status:** FAIL
- **Evidence:** `aide_integrity_checking.scheduled_checking` is null.
- **Recommendation:** Install AIDE and configure a cron job or systemd timer to regularly run AIDE checks (e.g., daily or weekly).

Rule 6.3.3: Ensure cryptographic mechanisms protect tools

- **Status:** FAIL
- **Evidence:** `aide_integrity_checking.audit_tools_integrity_tracked` is null.
- **Recommendation:** Install AIDE and configure its database to include critical system binaries and configuration files, ensuring they are checked with strong cryptographic hashes (e.g., `sha512`).

LOCAL USER Rules

Rule 7.2.1: Ensure accounts in `/etc/passwd` use shadowed passwords

- **Status:** PASS
- **Evidence:** The existence of `/etc/shadow` with appropriate permissions and the `pam_unix.so` obscure `yescrypt` configuration indicate shadowed passwords are in use.
- **Recommendation:** None

Rule 7.2.2: Ensure `/etc/shadow` password fields are not empty

- **Status:** PASS
- **Evidence:** `user_accounts.shadow_password_suite.user_chage_info` shows `last_password_change` dates for all users, implying non-empty password fields.
- **Recommendation:** None

Rule 7.2.3: Ensure all groups in `/etc/passwd` exist in `/etc/group`

- **Status:** PASS
- **Evidence:**
`user_accounts.local_users_and_groups_evidence.orphan_gids_in_passwd` is empty.
- **Recommendation:** None

Rule 7.2.4: Ensure shadow group is empty

- **Status:** PASS
- **Evidence:**
`user_accounts.local_users_and_groups_evidence.shadow_group_members` is empty.
- **Recommendation:** None

Rule 7.2.5: Ensure no duplicate UIDs exist

- **Status:** PASS
- **Evidence:**
`user_accounts.local_users_and_groups_evidence.duplicate_uids` is empty.
- **Recommendation:** None

Rule 7.2.6: Ensure no duplicate GIDs exist

- **Status:** PASS
- **Evidence:**
`user_accounts.local_users_and_groups_evidence.duplicate_gids` is empty.
- **Recommendation:** None

Rule 7.2.7: Ensure no duplicate user names exist

- **Status:** PASS
- **Evidence:**
`user_accounts.local_users_and_groups_evidence.duplicate_usernames` is empty.
- **Recommendation:** None

Rule 7.2.8: Ensure no duplicate group names exist

- **Status:** PASS
- **Evidence:**
`user_accounts.local_users_and_groups_evidence.duplicate_groupnames` is empty.
- **Recommendation:** None

Rule 7.2.9: Ensure local interactive user home directories are configured

- **Status:** PASS
- **Evidence:**
`user_accounts.local_users_and_groups_evidence.home_directory_issues` is empty.
- **Recommendation:** None

Rule 7.2.10: Ensure local interactive user dot files access is configured

- **Status:** PASS
- **Evidence:**
`user_accounts.local_users_and_groups_evidence.dotfile_permission_is` is empty.
- **Recommendation:** None

Section 3 – Network Configuration

Rule 3.1.1: IPv6 status identified

- **Status:** INFORMATIONAL
- **Evidence:**
 - `sysfs_disable_value: "0"` (IPv6 is enabled via sysfs)
 - `grub_ipv6_disable_flag: false` (IPv6 is not disabled in GRUB kernel command line)
 - `inet6_address_count: 2` (IPv6 addresses are active on at least one interface)
- **Recommendation:** None

Rule 3.1.2: Ensure wireless interfaces are not available

- **Status:** FAIL
- **Evidence:**
`network_config.network_devices.wireless_interfaces.any_wireless_fou` is true.
- **Recommendation:** If wireless is not required, disable or remove wireless hardware. Blacklist wireless kernel modules (e.g., `cfg80211`) using `modprobe -r <module_name>` and add `blacklist <module_name>` to `/etc/modprobe.d/blacklist.conf`.

Rule 3.1.3: Ensure Bluetooth services are not in use

- **Status:** PASS

- **Evidence:** `network_config.network_devices.bluetooth.bluez_installed` is false.
- **Recommendation:** None

Rule 3.2.1: Ensure the ATM kernel module is disabled

- **Status:** FAIL
- **Evidence:** `network_config.kernel_modules.atm.currently_loaded` is false, but `network_config.kernel_modules.atm.modprobe_config_lines` is empty, meaning it's not explicitly blacklisted or disabled.
- **Recommendation:** Add `install atm /bin/false` and `blacklist atm` to a file in `/etc/modprobe.d/`.

Rule 3.2.2: Ensure the CAN kernel module is disabled

- **Status:** FAIL
- **Evidence:** `network_config.kernel_modules.can.currently_loaded` is false, but `network_config.kernel_modules.can.modprobe_config_lines` is empty, meaning it's not explicitly blacklisted or disabled.
- **Recommendation:** Add `install can /bin/false` and `blacklist can` to a file in `/etc/modprobe.d/`.

Rule 3.2.3: Ensure the DCCP kernel module is disabled

- **Status:** FAIL
- **Evidence:** `network_config.kernel_modules.dccp.currently_loaded` is false, but `network_config.kernel_modules.dccp.modprobe_config_lines` is empty, meaning it's not explicitly blacklisted or disabled.
- **Recommendation:** Add `install dccp /bin/false` and `blacklist dccp` to a file in `/etc/modprobe.d/`.

Rule 3.2.4: Ensure the RDS kernel module is disabled

- **Status:** FAIL
- **Evidence:** `network_config.kernel_modules.rds.currently_loaded` is false, but `network_config.kernel_modules.rds.modprobe_config_lines` is empty, meaning it's not explicitly blacklisted or disabled.
- **Recommendation:** Add `install rds /bin/false` and `blacklist rds` to a file in `/etc/modprobe.d/`.

Rule 3.2.5: Ensure the SCTP kernel module is disabled

- **Status:** FAIL
- **Evidence:** `network_config.kernel_modules.sctp.currently_loaded` is false, but `network_config.kernel_modules.sctp.modprobe_config_lines` is empty, meaning it's not explicitly blacklisted or disabled.
- **Recommendation:** Add `install sctp /bin/false` and `blacklist sctp` to a file in `/etc/modprobe.d/`.

Rule 3.2.6: Ensure the TIPC kernel module is disabled

- **Status:** FAIL

- **Evidence:** `network_config.kernel_modules.tipc.currently_loaded` is `false`, but `network_config.kernel_modules.tipc.modprobe_config_lines` is empty, meaning it's not explicitly blacklisted or disabled.
- **Recommendation:** Add `install tipc /bin/false` and `blacklist tipc` to a file in `/etc/modprobe.d/`.

Rule 3.3.1.1: Ensure IPv4 packet forwarding is disabled

- **Status:** FAIL
- **Evidence:** `net.ipv4.ip_forward` runtime value is `"0"`, but `persisted_value` is `null`.
- **Recommendation:** Add `net.ipv4.ip_forward = 0` to a file in `/etc/sysctl.d/` (e.g., `/etc/sysctl.d/99-sysctl.conf`) and run `sudo sysctl -p`.

Rule 3.3.1.2: Ensure IPv4 forwarding is disabled on all interfaces

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.all.forwarding` runtime value is `"0"`, but `persisted_value` is `null`.
- **Recommendation:** Add `net.ipv4.conf.all.forwarding = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.3: Ensure IPv4 forwarding is disabled by default

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.forwarding` runtime value is `"0"`, but `persisted_value` is `null`.
- **Recommendation:** Add `net.ipv4.conf.default.forwarding = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.4: Ensure IPv4 ICMP redirects are not accepted

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.all.send_redirects` runtime value is `"1"`, expected `"0"`. `persisted_value` is `null`.
- **Recommendation:** Add `net.ipv4.conf.all.send_redirects = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.5: Ensure IPv4 ICMP redirects are not accepted by default

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.send_redirects` runtime value is `"1"`, expected `"0"`. `persisted_value` is `null`.
- **Recommendation:** Add `net.ipv4.conf.default.send_redirects = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.6: Ensure IPv4 bogus ICMP error responses are ignored

- **Status:** FAIL
- **Evidence:** `net.ipv4.icmp_ignore_bogus_error_responses` runtime value is `"1"`, but `persisted_value` is `null`.
- **Recommendation:** Add `net.ipv4.icmp_ignore_bogus_error_responses = 1`

to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.7: Ensure IPv4 broadcast ICMP requests are ignored

- **Status:** FAIL
- **Evidence:** `net.ipv4.icmp_echo_ignore_broadcasts` runtime value is "1", but `persisted_value` is null.
- **Recommendation:** Add `net.ipv4.icmp_echo_ignore_broadcasts = 1` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.8: Ensure IPv4 ICMP redirects are not accepted

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.all.accept_redirects` runtime value is "1", expected "0". `persisted_value` is null.
- **Recommendation:** Add `net.ipv4.conf.all.accept_redirects = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.9: Ensure IPv4 ICMP redirects are not accepted by default

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.accept_redirects` runtime value is "1", expected "0". `persisted_value` is null.
- **Recommendation:** Add `net.ipv4.conf.default.accept_redirects = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.10: Ensure IPv4 secure ICMP redirects are not accepted

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.all.secure_redirects` runtime value is "1", expected "0". `persisted_value` is null.
- **Recommendation:** Add `net.ipv4.conf.all.secure_redirects = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.11: Ensure IPv4 secure ICMP redirects are not accepted by default

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.secure_redirects` runtime value is "1", expected "0". `persisted_value` is null.
- **Recommendation:** Add `net.ipv4.conf.default.secure_redirects = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.12: Ensure IPv4 reverse path filtering is enabled

- **Status:** PASS
- **Evidence:** `net.ipv4.conf.all.rp_filter` runtime value is "2" and `persisted_value` is "2".
- **Recommendation:** None

Rule 3.3.1.13: Ensure IPv4 reverse path filtering is enabled by default

- **Status:** PASS
- **Evidence:** `net.ipv4.conf.default.rp_filter` runtime value is "2" and `persisted_value` is "2".
- **Recommendation:** None

Rule 3.3.1.14: Ensure IPv4 source routed packets are not accepted

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.all.accept_source_route` runtime value is "0", but `persisted_value` is null.
- **Recommendation:** Add `net.ipv4.conf.all.accept_source_route = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.15: Ensure IPv4 source routed packets are not accepted by default

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.accept_source_route` runtime value is "1", expected "0". `persisted_value` is null.
- **Recommendation:** Add `net.ipv4.conf.default.accept_source_route = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.16: Ensure IPv4 martian packets are logged

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.all.log_martians` runtime value is "0", expected "1". `persisted_value` is null.
- **Recommendation:** Add `net.ipv4.conf.all.log_martians = 1` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.17: Ensure IPv4 martian packets are logged by default

- **Status:** FAIL
- **Evidence:** `net.ipv4.conf.default.log_martians` runtime value is "0", expected "1". `persisted_value` is null.
- **Recommendation:** Add `net.ipv4.conf.default.log_martians = 1` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.1.18: Ensure TCP SYN Cookies is enabled

- **Status:** FAIL
- **Evidence:** `net.ipv4.tcp_syncookies` runtime value is "1", but `persisted_value` is null.
- **Recommendation:** Add `net.ipv4.tcp_syncookies = 1` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.2.1: Ensure IPv6 packet forwarding is disabled

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.all.forwarding` runtime value is "0", but `persisted_value` is null.
- **Recommendation:** Add `net.ipv6.conf.all.forwarding = 0` to a file in `/etc/`

`sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.2.2: Ensure IPv6 forwarding is disabled by default

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.default.forwarding` runtime value is "0", but `persisted_value` is null.
- **Recommendation:** Add `net.ipv6.conf.default.forwarding = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.2.3: Ensure IPv6 ICMP redirects are not accepted

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.all.accept_redirects` runtime value is "1", expected "0". `persisted_value` is null.
- **Recommendation:** Add `net.ipv6.conf.all.accept_redirects = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.2.4: Ensure IPv6 ICMP redirects are not accepted by default

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.default.accept_redirects` runtime value is "1", expected "0". `persisted_value` is null.
- **Recommendation:** Add `net.ipv6.conf.default.accept_redirects = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.2.5: Ensure IPv6 source routed packets are not accepted

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.all.accept_source_route` runtime value is "0", but `persisted_value` is null.
- **Recommendation:** Add `net.ipv6.conf.all.accept_source_route = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.2.6: Ensure IPv6 source routed packets are not accepted by default

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.default.accept_source_route` runtime value is "0", but `persisted_value` is null.
- **Recommendation:** Add `net.ipv6.conf.default.accept_source_route = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.2.7: Ensure IPv6 router advertisements are not accepted

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.all.accept_ra` runtime value is "1", expected "0". `persisted_value` is null.
- **Recommendation:** Add `net.ipv6.conf.all.accept_ra = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Rule 3.3.2.8: Ensure IPv6 router advertisements are not accepted by

default

- **Status:** FAIL
- **Evidence:** `net.ipv6.conf.default.accept_ra` runtime value is "1", expected "0". `persisted_value` is null.
- **Recommendation:** Add `net.ipv6.conf.default.accept_ra = 0` to a file in `/etc/sysctl.d/` and run `sudo sysctl -p`.

Section 5.3 – Pluggable Authentication Modules (PAM)

Rule 5.3.1.1: Ensure libpam-runtime is installed

- **Status:** PASS
- **Evidence:** `pam.pam_packages.libpam-runtime.installed` is true and `installed_version` matches `candidate_version` (1.5.3-5ubuntu5.5).
- **Recommendation:** None

Rule 5.3.1.2: Ensure libpam-modules is installed

- **Status:** PASS
- **Evidence:** `pam.pam_packages.libpam-modules.installed` is true and `installed_version` matches `candidate_version` (1.5.3-5ubuntu5.5).
- **Recommendation:** None

Rule 5.3.1.3: Ensure libpam-pwquality is installed

- **Status:** FAIL
- **Evidence:** `pam.pam_packages.libpam-pwquality.installed` is false.
- **Recommendation:** Install the `libpam-pwquality` package: `sudo apt install libpam-pwquality`.

Rule 5.3.1.4: Ensure cracklib-runtime is installed

- **Status:** FAIL
- **Evidence:** `pam.pam_packages.cracklib-runtime.installed` is false.
- **Recommendation:** Install the `cracklib-runtime` package: `sudo apt install cracklib-runtime`.

Rule 5.3.2.1: Ensure pam_unix is enabled

- **Status:** PASS
- **Evidence:** `pam.pam_profiles_raw.module_lines.pam_unix.so` contains active lines in `common-auth`, `common-account`, `common-password`, and `common-session`.
- **Recommendation:** None

Rule 5.3.2.2: Ensure pam_faillock is enabled

- **Status:** FAIL
- **Evidence:** `pam.pam_profiles_raw.module_lines.pam_faillock.so` is empty, indicating `pam_faillock` is not enabled.

- **Recommendation:** Add `auth required pam_faillock.so preauth audit deny=5 unlock_time=900` and `auth [default=die] pam_faillock.so authfail audit deny=5 unlock_time=900` to the `auth` section of `/etc/pam.d/common-auth` (or relevant service files) and `account required pam_faillock.so` to the `account` section of `/etc/pam.d/common-account`.

Rule 5.3.2.3: Ensure pam_pwquality is enabled

- **Status:** FAIL
- **Evidence:** `pam.pam_profiles_raw.module_lines.pam_pwquality.so` is empty, indicating `pam_pwquality` is not enabled.
- **Recommendation:** Install `libpam-pwquality` and add `password requisite pam_pwquality.so retry=3` to the `password` section of `/etc/pam.d/common-password`.

Rule 5.3.2.4: Ensure pam_pwhistory is enabled

- **Status:** FAIL
- **Evidence:** `pam.pam_profiles_raw.module_lines.pam_pwhistory.so` is empty, indicating `pam_pwhistory` is not enabled.
- **Recommendation:** Add `password requisite pam_pwhistory.so remember=24 use_authok` to the `password` section of `/etc/pam.d/common-password`.

Rule 5.3.3.1.1: Ensure pam_faillock deny is configured to 5 or less

- **Status:** FAIL
- **Evidence:** `pam.pam_faillock.faillock_conf_settings.deny` is null and `pam.pam_faillock.faillock_inline_pam_settings.deny` is null.
- **Recommendation:** Configure `pam_faillock` with `deny=5` in `/etc/security/faillock.conf` or as an inline argument in `/etc/pam.d/common-auth`.

Rule 5.3.3.1.2: Ensure pam_faillock unlock_time is configured to 900 or greater

- **Status:** FAIL
- **Evidence:** `pam.pam_faillock.faillock_conf_settings.unlock_time` is null and `pam.pam_faillock.faillock_inline_pam_settings.unlock_time` is null.
- **Recommendation:** Configure `pam_faillock` with `unlock_time=900` (or 0 for never unlock) in `/etc/security/faillock.conf` or as an inline argument in `/etc/pam.d/common-auth`.

Rule 5.3.3.1.3: Ensure pam_faillock even_deny_root / root_unlock_time is configured

- **Status:** FAIL
- **Evidence:** `pam.pam_faillock.faillock_conf_settings.even_deny_root` is null and `pam.pam_faillock.faillock_inline_pam_settings.even_deny_root` is null.
- **Recommendation:** Configure `pam_faillock` with `even_deny_root` and

root_unlock_time=60 (or greater) in /etc/security/faillock.conf or as inline arguments in /etc/pam.d/common-auth.

Rule 5.3.3.2.1: Ensure pam_pwquality difok is configured to 2 or greater

- **Status:** FAIL
- **Evidence:** pam.pam_pwquality.pwquality_conf_settings.difok is null and pam.pam_pwquality.pwquality_inline_pam_settings.difok is null.
- **Recommendation:** Configure pam_pwquality with difok=2 in /etc/security/pwquality.conf or as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.2.2: Ensure pam_pwquality minlen is configured to 14 or greater

- **Status:** FAIL
- **Evidence:** pam.pam_pwquality.pwquality_conf_settings.minlen is null and pam.pam_pwquality.pwquality_inline_pam_settings.minlen is null.
- **Recommendation:** Configure pam_pwquality with minlen=14 in /etc/security/pwquality.conf or as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.2.3: (Manual) Password complexity (dcredit, ucredit, lcredit, ocredit)

- **Status:** INFORMATIONAL
- **Evidence:** pam.pam_pwquality.pwquality_conf_settings and pam.pam_pwquality.pwquality_inline_pam_settings show dcredit, ucredit, lcredit, ocredit are all null.
- **Recommendation:** Manually review and configure password complexity requirements (e.g., dcredit=-1, ucredit=-1, lcredit=-1, ocredit=-1 for one digit, uppercase, lowercase, and other character respectively) in /etc/security/pwquality.conf or as inline arguments in /etc/pam.d/common-password.

Rule 5.3.3.2.4: Ensure pam_pwquality maxrepeat is configured to 3 or less

- **Status:** FAIL
- **Evidence:** pam.pam_pwquality.pwquality_conf_settings.maxrepeat is null and pam.pam_pwquality.pwquality_inline_pam_settings.maxrepeat is null.
- **Recommendation:** Configure pam_pwquality with maxrepeat=3 in /etc/security/pwquality.conf or as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.2.5: Ensure pam_pwquality maxsequence is configured to 3 or less

- **Status:** FAIL
- **Evidence:** pam.pam_pwquality.pwquality_conf_settings.maxsequence is null and pam.pam_pwquality.pwquality_inline_pam_settings.maxsequence is null.
- **Recommendation:** Configure pam_pwquality with maxsequence=3 in /etc/

security/pwquality.conf or as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.2.6: Ensure pam_pwquality dictcheck is configured to 1

- **Status:** FAIL
- **Evidence:** pam.pam_pwquality.pwquality_conf_settings.dictcheck is null and pam.pam_pwquality.pwquality_inline_pam_settings.dictcheck is null.
- **Recommendation:** Configure pam_pwquality with dictcheck=1 in /etc/security/pwquality.conf or as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.2.7: Ensure pam_pwquality enforcing is configured to 1

- **Status:** FAIL
- **Evidence:** pam.pam_pwquality.pwquality_conf_settings.enforcing is null and pam.pam_pwquality.pwquality_inline_pam_settings.enforcing is null.
- **Recommendation:** Configure pam_pwquality with enforcing=1 in /etc/security/pwquality.conf or as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.2.8: Ensure pam_pwquality enforce_for_root is present

- **Status:** FAIL
- **Evidence:**
pam.pam_pwquality.pwquality_conf_settings.enforce_for_root is null and
pam.pam_pwquality.pwquality_inline_pam_settings.enforce_for_root is null.
- **Recommendation:** Configure pam_pwquality with enforce_for_root in /etc/security/pwquality.conf or as an inline argument in /etc/pam.d/common-password.

Rule 5.3.3.3.1: Ensure pam_pwhistory remember is configured to 24 or greater

- **Status:** FAIL
- **Evidence:** pam.pam_pwhistory.pwhistory_inline_pam_settings.remember is null.
- **Recommendation:** Configure pam_pwhistory with remember=24 in /etc/pam.d/common-password.

Rule 5.3.3.3.2: Ensure pam_pwhistory enforce_for_root is present

- **Status:** FAIL
- **Evidence:**
pam.pam_pwhistory.pwhistory_inline_pam_settings.enforce_for_root is null.
- **Recommendation:** Configure pam_pwhistory with enforce_for_root in /etc/pam.d/common-password.

Rule 5.3.3.3.3: Ensure pam_pwhistory use_authtok is present

- **Status:** FAIL
- **Evidence:**
`pam.pam_pwhistory.pwhistory_inline_pam_settings.use_authtok` is null.
- **Recommendation:** Configure `pam_pwhistory` with `use_authtok` in `/etc/pam.d/common-password`.

Rule 5.3.3.4.1: Ensure pam_unix nullok is absent

- **Status:** FAIL
- **Evidence:** `pam.pam_unix.pam_unix_flag_presence.nullok` is "present" in `/etc/pam.d/common-auth`.
- **Recommendation:** Remove the `nullok` option from the `pam_unix.so` line in `/etc/pam.d/common-auth` to prevent users from logging in with empty passwords.

Rule 5.3.3.4.2: Ensure pam_unix remember is absent

- **Status:** PASS
- **Evidence:** `pam.pam_unix.pam_unix_flag_presence.remember` is null.
- **Recommendation:** None

Rule 5.3.3.4.3: Ensure strong password hashing algorithm is configured

- **Status:** PASS
- **Evidence:** `pam.pam_unix.pam_unix_flag_presence.yescrypt` is "present".
- **Recommendation:** None

Rule 5.3.3.4.4: Ensure pam_unix use_authtok is present (in password stack)

- **Status:** FAIL
- **Evidence:** `pam.pam_unix.pam_unix_flag_presence.use_authtok` is null.
- **Recommendation:** Add the `use_authtok` option to the `pam_unix.so` line in the password stack of `/etc/pam.d/common-password`.

SUID/SGID Triage Report

```
{
  "findings": [],
  "summary": {
    "suspicious_found": 0,
    "message": "No unexpected SUID/SGID files detected."
  }
}
```